

Exp 11:

JOINS

Sample records in Department, Faculty, and Student tables:

```
mysql> SELECT * FROM Department;
+-----+-----+
| Department_ID | Department_Name |
+-----+-----+
|          1   | Sales           |
|          2   | IT              |
|          3   | HR              |
+-----+-----+
3 rows in set (0.01 sec)
```

```
mysql> SELECT * FROM Faculty;
+-----+-----+-----+
| Faculty_ID | Faculty_Name | Department_ID |
+-----+-----+-----+
|        101 | Ratnam       |             1 |
|        102 | Kumar        |             2 |
|        103 | Priya        |             3 |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Student;
+-----+-----+
| Student_ID | Student_Name |
+-----+-----+
|          1 | Arun         |
|          2 | Priya        |
|          3 | Rahul        |
+-----+-----+
3 rows in set (0.00 sec)
```

Sample records in Course, Score, and Teaches tables:

```
mysql> SELECT * FROM Course;
+-----+-----+
| Course_No | Course_Name |
+-----+-----+
| C101      | DBMS        |
| C102      | Python      |
| C103      | Java        |
+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Score;
+-----+-----+-----+
| Student_ID | Course_No | Score |
+-----+-----+-----+
|          1 | C101      |    90 |
|          1 | C102      |    85 |
|          2 | C101      |    80 |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Teaches;
+-----+-----+
| Faculty_ID | Course_No |
+-----+-----+
|          101 | C101      |
|          101 | C102      |
|          102 | C103      |
+-----+-----+
3 rows in set (0.00 sec)
```

List the departments where the faculty members are working

```
mysql> SELECT DISTINCT d.Department_Name
-> FROM Department d
-> INNER JOIN Faculty f ON d.Department_ID = f.Department_ID;
+-----+
| Department_Name |
+-----+
| Sales           |
| IT              |
| HR              |
+-----+
3 rows in set (0.00 sec)
```

Find the student who has no score in any of the courses. List the student name and course number.

```
mysql> SELECT s.Student_Name, c.Course_No
-> FROM Student s
-> CROSS JOIN Course c
-> LEFT JOIN Score sc ON s.Student_ID = sc.Student_ID AND c.Course_No = sc.Course_No
-> WHERE sc.Student_ID IS NULL;
+-----+-----+
| Student_Name | Course_No |
+-----+-----+
| Rahul        | C101      |
| Rahul        | C102      |
| Rahul        | C103      |
+-----+-----+
3 rows in set (0.00 sec)
```

The office clerk needs the names of the courses taken by the faculty belonging to the 'Sales' department whose name is 'Ratnam'

```
mysql> SELECT c.Course_Name
-> FROM Faculty f
-> INNER JOIN Department d ON f.Department_ID = d.Department_ID
-> INNER JOIN Teaches t ON f.Faculty_ID = t.Faculty_ID
-> INNER JOIN Course c ON t.Course_No = c.Course_No
-> WHERE d.Department_Name = 'Sales' AND f.Faculty_Name = 'Ratnam';
+-----+
| Course_Name |
+-----+
| DBMS        |
| Python      |
+-----+
2 rows in set (0.00 sec)
```